

HCSM-00

Foundations, Notation, and Mapping The Hodge Complex Standard Model: Substrate, Hodge Complex, Spectral Midpoint, Processor, Output Structures, Emergent Physics, and Cosmology

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by two independent anchors (photon helicity for $d = 4$; equal entropy spacing for $L = 8$) and one derived anchor (the topology of the Hodge-Dirac kernel). The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework's foundational objects and states the notation. All seven rank blockers of the framework's derivation program are closed. The framework has three discrete disclosed inputs (Standard Model content; physical identifications; identification thesis) and one dimensionful anchor (the Planck mass M_P). Every other quantity is derived from the substrate.

Session update (September 2026, continued session 2). Five new structural results are integrated: (i) the *Kernel-Diagonal Theorem* (HCSM-60), which proves that the diagonal of the projector onto $\ker(H_{\text{wind}})$ takes exactly two values ($2/d$ on $\mathcal{O}_0$, $(d-1)/d$ on $\mathcal{O}_1 \cup \mathcal{O}_2$) and yields the EFT coefficient $c = 5/9$, the mass-map coefficient $(d-1)/2 = 3/2$, and the kernel dimension $10 = (d-1)^2 + 1$ as three traces of one projector; (ii) the *absence of a light dark photon*, $\epsilon \equiv 0$ exactly (HCSM-41); (iii) the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ grading of the 56-dimensional Hodge lift (HCSM-42); (iv) the *strong-CP solution*, $\theta_{\text{QCD}} = 0$ exactly, from the orientation- $\mathbb{Z}_2$ anticommutation $\{F, H_{\text{wind}}\} = 0$ (HCSM-43); (v) the Proca sector and the $\mathfrak{u}(1)_Y$ origin, with $m_W/m_Z = \cos \theta_W$ (HCSM-44). The Λ -promotion mechanism (item 6.13) is closed: the residual vacuum energy is promoted by pure projection, and the hidden $\mathfrak{u}(1)$ VEVs vanish. The refined dark energy density matches observation to 0.010%.

Contents

1	Introduction	3
1.1	What HCSM is	3
1.2	The single input $d = 4$	3

1.3	Architectural levels	3
1.4	Scope and honesty	4
1.5	Continuum limit and emergent Lorentz invariance	4
2	Level 0 — The Substrate	4
2.1	The periodic 8×8 torus	4
2.2	The discrete 5-point Laplacian	4
2.3	The two independent anchors and the derived third anchor	4
2.4	The size relation $N = d \cdot 2^d$	5
2.5	The 14-mode count at the self-paired eigenvalue	5
3	Level 1 — The Hodge Complex	5
3.1	Differential forms on the torus	5
3.2	Exterior derivative and codifferential	5
3.3	The Hodge-Dirac operator	5
3.4	The $\mathbb{Z}_2$ grading	5
3.5	The kernel and the emergence of four dimensions	5
4	Level 2 — The Spectral Midpoint	5
4.1	The self-paired eigenvalue	5
4.2	The three D_4 orbits	5
4.3	The Hodge lift to the 56-dimensional sector	6
5	Level 3 — The Processor and Its Dynamics	6
5.1	The winding generator	6
5.2	The three winding classes	6
5.3	The processor T	6
5.4	F/B dimensions and internal ratios	6
5.5	The Landauer channel	6
5.6	The zero state	6
5.7	Mirror and diagonal-flip maps	7
5.8	The five-layer loop	7
5.9	The exact ratio and the loop-pass duration	7
5.10	The Kernel-Diagonal Theorem (HCSM-60)	7
5.11	F-Invariance of the Winding Kernel (HCSM-60)	7
5.12	Kernel- T Decomposition (HCSM-60)	7
6	Level 4 — Output Structures	8
6.1	Charge	8
6.2	Mass	8
7	Level 5 — Emergent Physics	8
7.1	Flavor: the CKM matrix	8
7.2	Flavor: the PMNS matrix	8
7.3	Higgs sector	8
7.4	Gravity	8
7.5	Absence of a Light Dark Photon (HCSM-41)	9
7.6	Spin Chirality $\mathbb{Z}_2$ Grading (HCSM-42)	9
7.7	Strong CP from Orientation $\mathbb{Z}_2$ (HCSM-43)	9

7.8 Proca Sector and $u(1)_Y$ Origin (HCSM-44)	9
8 Level 6 — Cosmology	9
8.1 The kL crossover	9
8.2 Dark energy and the DME entropy-well normalization	9
8.3 Vacuum stability	10
8.4 The Λ -promotion mechanism	10
9 Session-Level Theorem Inventory	10
10 Reproducibility	11
11 Conclusion	11

1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model is a structural derivation of the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature, but the residue left behind when a small, discrete, information-bearing lattice cancels itself against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The framework's answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content.

1.2 The single input $d = 4$

The whole structure of HCSM traces back to a single input: the spacetime dimension $d = 4$. This input is itself derived from the photon's two helicity states, and the substrate size $L = 8$ is selected independently by the equal-entropy-spacing theorem on the torus Laplacian. The two derivations agree, and together they force the substrate and the emergent dimension.

1.3 Architectural levels

The framework's architecture is a chain from the substrate to observable physics, organized in six levels.

Level 0. Substrate. The periodic 8×8 torus Λ , the discrete 5-point Laplacian, the two independent anchors.

Level 1. Hodge complex. The differential forms $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, the Hodge-Dirac operator $D = d + d^*$, the $\mathbb{Z}_2$ grading, the kernel $\ker D \cong H^0 \oplus H^1 \oplus H^2$.

Level 2. Spectral midpoint. The self-paired eigenvalue $\lambda = 4$, its multiplicity 14, the three D_4 orbits, the Hodge lift to the 56-dimensional sector.

Level 3. Processor and dynamics. The winding generator H_{wind} , the three winding classes, the processor $T = I - P_C$, the fermion and boson sectors, the Landauer channel, the loop operator, and the *Kernel-Diagonal Theorem* that unifies the framework's three leading coefficients.

Level 4. Output structures. The charge functional q_3 , the mass map, the $N/4$ labels.

Level 5. Emergent physics. Flavor (CKM, PMNS), the Higgs sector, gravity, the Newton constant, the gauge content including the absence of a dark photon, the spin-chirality grading, the strong-CP solution, and the Proca sector.

Level 6. Cosmology. The kL crossover, dark energy, the DME entropy-well normalization, vacuum stability, and the physical mechanism by which the residual vacuum energy is promoted to the observed cosmological constant.

1.4 Scope and honesty

Every theorem stated as HCSM-native uses only the following primitives: the periodic 8×8 torus Λ ; the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$; the Hodge-Dirac operator $D = d + d^*$; the D_4 point-group action on Λ ; the discrete 5-point Laplacian Δ ; the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$; the zero-state vector s ; the charge functional $q_3 = F(n_1, \sigma)$; the $N/4$ labels; the processor $T = I - P_C$ and its class partition.

No imported machinery appears in any HCSM-native theorem. Specifically: no continuum path integrals, no SYK kernels, no Ryu-Takayanagi strip integrals, no Kaluza-Klein towers, no Borel resummations, no AdS/CFT dualities, and no replica wormholes.

The framework's disclosed-input set consists of three discrete items (Standard Model per-generation content; physical identifications; the identification thesis 14 modes $\leftrightarrow$ 14 SM particles), extended in this session by four additional physical identifications: $A_1 \leftrightarrow$ color and $B_1 \leftrightarrow$ weak isospin (prior); $F \leftrightarrow$ CP, $\chi_{\text{spin}} \leftrightarrow$ L/R chirality, and $H_{\text{wind}} \leftrightarrow F_{\mu\nu} \tilde{F}^{\mu\nu}$ (this session). The dimensionful anchor is the Planck mass M_P . No second dimensionful anchor is required; the DME entropy-well normalization S_0 is derived from M_P and HCSM-native dimensionless factors.

1.5 Continuum limit and emergent Lorentz invariance

The substrate is not a discretization of an underlying continuum. The framework's claim is the opposite: the continuum emerges from the substrate. The physical scale is set by the mass map and the kL crossover. The discrete 5-point Laplacian has the expansion $\lambda(n_1, n_2) = (\pi^2/16)(n_1^2 + n_2^2) + O(n^4)$, isotropic to leading order. The quartic corrections are suppressed by ϵ^2 relative to the leading term.

2 Level 0 — The Substrate

2.1 The periodic 8×8 torus

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, $N = 64$ sites, with periodic boundary conditions.

2.2 The discrete 5-point Laplacian

$(\Delta f)(x, y) = 4f(x, y) - f(x+1, y) - f(x-1, y) - f(x, y+1) - f(x, y-1)$, with eigenvalues $\lambda(n_1, n_2) = 4 - 2\cos(\pi n_1/4) - 2\cos(\pi n_2/4)$.

2.3 The two independent anchors and the derived third anchor

Anchor 1 (input): photon helicity. A massless spin-1 particle has two helicity states; two states $\times$ two real components = four real numbers; Wick rotation gives $d = 4$.

Anchor 2 (input): equal entropy spacing. Among all integers $4 \leq L \leq 10^4$, only $L = 8$ gives three equally spaced entropy classes. Hence $L = 8$.

Anchor 3 (derived): kernel topology. The kernel of D has dimension $b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$, giving the emergent 4D spacetime.

2.4 The size relation $N = d \cdot 2^d$

Theorem 1 (Substrate size). $N = d \cdot 2^d$.

At $d = 4$: $N = 64$, a perfect square, forcing $L = \sqrt{N} = 8$.

2.5 The 14-mode count at the self-paired eigenvalue

$\text{mult}_{\lambda=4} = 2(L - 1) = 14$. This matches one Standard Model generation.

3 Level 1 — The Hodge Complex

3.1 Differential forms on the torus

Ω^0 (dim 64, bosonic), Ω^1 (dim 128, fermionic), Ω^2 (dim 64, bosonic). Total dim = 256.

3.2 Exterior derivative and codifferential

$d_0 f(x, \mu) = f(x + e_\mu) - f(x)$; $d_1 \omega(x) = \omega_2(x + e_1) - \omega_2(x) - \omega_1(x + e_2) + \omega_1(x)$; $d_1 d_0 = 0$ exactly.

3.3 The Hodge-Dirac operator

$D = d + d^*$, with $D^2|_{\Omega^k} = \Delta$ exactly on each form degree.

3.4 The $\mathbb{Z}_2$ grading

Even forms ($\Omega^0 \oplus \Omega^2$) bosonic; odd forms (Ω^1) fermionic. D is odd with respect to this grading.

3.5 The kernel and the emergence of four dimensions

$\ker D \cong H^0 \oplus H^1 \oplus H^2$, $\dim \ker D = 4$. The four harmonic forms are the four spacetime directions. The metric on $\ker D$ has signature $(3, 1)$; the isometry group is exactly $SO(3, 1)$.

4 Level 2 — The Spectral Midpoint

4.1 The self-paired eigenvalue

$\lambda = 4$; multiplicity 14.

4.2 The three D_4 orbits

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\},$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\},$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}.$$

4.3 The Hodge lift to the 56-dimensional sector

$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$. D_4 decomposition: $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$.

The 56-dimensional Hodge lift carries a $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ grading generated by three commuting involutions (HCSM-42):

- $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$: Hodge form-parity.
- χ_{spin} : spin-structure parity (distinguishes even spin structures (NS, NS) , (R, R) from odd (NS, R) , (R, NS)).
- $F = (n_1, n_2) \rightarrow (n_2, n_1)$: orientation flip.

All three square to I and pairwise commute. Consequences: spin-statistics from γ_5 with $\{\gamma_5, D\} = 0$ and $\text{Tr}(\gamma_5 f(D)) = 0$ for any even regulator f (substrate-level two-loop anomaly cancellation); chirality from χ_{spin} ; CP from F .

5 Level 3 — The Processor and Its Dynamics

5.1 The winding generator

$H_{\text{wind}} = i \sin(\pi M/8)$, $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$, where $\sin$ is applied elementwise. Hermitian, entries in $\mathbb{Q}(\sqrt{2})$. Characteristic polynomial $p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32)$; spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.

5.2 The three winding classes

$A = \{\nu, e, t\}$ (size 3); $B = \{u, d, s, \mu, c, \tau, b\}$ (size 7); $C = \{\text{dark}, W, Z, H\}$ (size 4). These classes are defined by HCSM-14; they are the winding classes of the framework, not the trajectory-frequency partition of the position-space wavefunctions. Class projectors P_A, P_B, P_C are diagonal, orthogonal, idempotent, and sum to the identity. The ratio $\omega_B/\omega_A = 3 = d - 1$.

5.3 The processor T

$T = P_{\text{mesh}} + P_\tau = I - P_C$, rank-10 orthogonal projection. Image is the fermion sector F (size 10); kernel is the boson sector B (size 4).

5.4 F/B dimensions and internal ratios

$$|F| = d(d+1)/2, |B| = d, (|F| - |B|)/|B| = (d-1)/2, |F|/|B| = (d+1)/2.$$

5.5 The Landauer channel

$$S_{\text{in}} = \log 56, S_{\text{out}} = \log 40, \Delta S = \log(7/5) = 0.33647 \dots \text{nats}.$$

5.6 The zero state

14×19 measurement matrix M ; rank 13, nullity 1; unique nullspace vector s . Odd under the diagonal reflection $(n_1, n_2) \leftrightarrow (n_2, n_1)$; the two fixed points $(2, 2)$ and $(6, 6)$ carry $s = 0$ exactly (the crossover zeros, coinciding with the observer pair $\{W, t\}$).

5.7 Mirror and diagonal-flip maps

$M : (n_1, n_2) \mapsto (4 - n_1, 4 - n_2)$ (mirror); $F : (n_1, n_2) \mapsto (n_2, n_1)$ (diagonal flip). On the 14-mode multiplet: $|\text{Fix}(M) \cap \mathcal{O}_2| = 4 = |\mathcal{O}_2|$; $\text{Fix}(M) \cap \text{Fix}(F) = \{(2, 2), (6, 6)\}$.

5.8 The five-layer loop

$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$, acting diagonally on the 14-mode multiplet.

5.9 The exact ratio and the loop-pass duration

$|M_A|/|M_B| = 3 = d - 1$; $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$ by Landauer selection.

5.10 The Kernel-Diagonal Theorem (HCSM-60)

Let P_{ker} be the orthogonal projector onto $\ker(H_{\text{wind}})$.

Theorem 1 (Kernel-Diagonal). *The diagonal of P_{ker} in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} \frac{2}{d} = \frac{1}{2}, & i \in \mathcal{O}_0, \\ \frac{d-1}{d} = \frac{3}{4}, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (1)$$

Proof sketch. By HCSM-14, $\{H_{\text{wind}}, P_F\} = 0$, so H_{wind}^2 commutes with P_F and $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$ is F -invariant. The diagonal of P_{ker} is constant on each F -cycle. Enumerating the six F -cycles (one is $\{\nu, e\} \subset \mathcal{O}_0$; five are in $\mathcal{O}_1 \cup \mathcal{O}_2$) gives the two values. Full proof in HCSM-60.

Corollary 2 (Three coefficients from one diagonal). *With P_{ker} as in Theorem 1,*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (\text{EFT coefficient}) \quad (2)$$

$$\frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (\text{mass-map coefficient}) \quad (3)$$

$$\dim \ker(H_{\text{wind}}) = 1 + (d-1)^2 = 10. \quad (\text{kernel dimension}) \quad (4)$$

The three coefficients are traces of the same diagonal: the partial trace on $\mathcal{O}_0$ gives $1 = d/d$; the partial trace on $\mathcal{O}_1 \cup \mathcal{O}_2$ gives $(d-1)^2 = 9$; the total trace gives 10.

5.11 F-Invariance of the Winding Kernel (HCSM-60)

Theorem 3 (F -invariance). $[P_{\text{ker}}, P_F] = 0$, while $[P_{\text{ker}}, R(r)] \neq 0$ and $[P_{\text{ker}}, R(s)] \neq 0$. At machine precision, $\|[P_{\text{ker}}, P_F]\|_F < 10^{-14}$, and $\|[P_{\text{ker}}, R(r)]\|_F = \|[P_{\text{ker}}, R(s)]\|_F = \sqrt{2}$.

The full D_4 point group does *not* preserve $\ker(H_{\text{wind}})$; only the diagonal flip F does. The two-value diagonal is a consequence of F -invariance, not of full D_4 symmetry.

5.12 Kernel- T Decomposition (HCSM-60)

Theorem 4 (Kernel splitting under the processor). $\dim(\ker(H_{\text{wind}}) \cap \text{im}(T)) = 6 = |F| - |B|$; $\dim(\ker(H_{\text{wind}}) \cap \ker(T)) = 4 = |B|$. The two dimensions sum to $10 = \dim \ker(H_{\text{wind}})$.

The kernel of the winding generator intersects the bosonic sector B in 4 dimensions and the fermionic sector F in 6 dimensions.

6 Level 4 — Output Structures

6.1 Charge

$q_3 = F(n_1, \sigma)$: bijection from the 14 realized pairs to the five SM charge eigenvalues $\{-3, -1, 0, +2, +3\}$.

6.2 Mass

$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$, with $A_{\text{mass}} = 224$, $B_{\text{mass}} = 0.5411967\dots$. The coefficient $(d - 1)/2 = 3/2$ has three independent derivations:

1. From the processor F/B split: $(|F| - |B|)/|B| = (d - 1)/2$.
2. From the DNLS Hartree orbit trace: $\text{Tr } K_{\mathcal{O}_1} / \text{Tr } K_{\mathcal{O}_0} = 3/2$, exact.
3. From the kernel-diagonal ratio (HCSM-60): $\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2} / \text{diag}(P_{\text{ker}})|_{\mathcal{O}_0} = (d - 1)/2$.

The third derivation is the most primitive: it uses only the projector onto $\ker(H_{\text{wind}})$, not the Hartree kernel or the processor's F/B split. The mass map reproduces the twelve charged fermion masses at RMS 0.0449 dex with zero free parameters.

7 Level 5 — Emergent Physics

7.1 Flavor: the CKM matrix

$\sin \theta_{12} = \pi/14 = 0.2243995\dots$. The winding amplitude $a = 3/5$ is *topological*: it is the ratio $(|F| - |B|)/|F| = 6/10$, a ratio of exact rank counts, and receives no δ_{edge} correction (HCSM-37, topological protection theorem). Wolfenstein parameters: $\lambda = \pi/14$, $A = (d/(d + 1))(1 + \delta_{\text{edge}})$, $\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = (d + 1)/(d(d - 1)) = 5/12$.

7.2 Flavor: the PMNS matrix

Neutrino $N/4$ spectrum (60, 60, 58); PMNS magnitudes at RMS 0.035; mixing angles $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$; Dirac CP phase $\delta_{CP} \approx 197^\circ$.

The generation phase $\phi_{\text{gen}} = (|F| - |B|)/|F| \cdot \pi = (3/5)\pi$ is *exactly* $(3/5)\pi$ by the topological protection theorem (HCSM-37). No $\delta_{\text{edge}}/6$ correction applies, because ϕ_{gen} has no physical-scale normalization.

Majorana phases: $(\alpha_1, \alpha_2) = (\pi, 0) \pmod{\pi}$. The Majorana sector is CP-conserving.

7.3 Higgs sector

$y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ with $4/3 = |\mathcal{O}_2|/(d - 1)$; $m_t = 172.688398 \text{ GeV}$ (PDG $172.690 \pm 0.30 \text{ GeV}$). $\lambda_{\text{tree}} = \ln(kL)/[(d - 1)\pi^2]$. $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$.

7.4 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes. Exactly massless at all orders by D_4 Schur orthogonality. $G_N = (d + 1)^2/(8\pi M_P^2) = 25/(8\pi M_P^2)$.

7.5 Absence of a Light Dark Photon (HCSM-41)

Theorem 5. *The HCSM kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly over all mass scales.*

The HCSM gauge algebra $\mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ contains no hidden Abelian factor coupling to electrically charged matter. The hidden sector $5 + 6 + 31$ is $U(1)_Y$ -neutral. Consequently: no tree-level kinetic mixing operator $F_{\mu\nu}F'^{\mu\nu}$ can be written; no one-loop kinetic mixing is generated; no state is charged under both the visible $\mathfrak{u}(1)_{\text{em}}$ and any hidden Abelian factor. The prediction is falsifiable by SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC at $\epsilon > 10^{-12}$.

7.6 Spin Chirality $\mathbb{Z}_2$ Grading (HCSM-42)

The 56-dimensional Hodge lift at $\lambda = 4$ carries three commuting $\mathbb{Z}_2$ operators: γ_5 (Hodge form-parity), χ_{spin} (spin-structure parity), and F (orientation). They generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Consequences: spin-statistics from γ_5 with $\{\gamma_5, D\} = 0$ and $\text{Tr}(\gamma_5 f(D)) = 0$ for any even regulator f ; chirality from χ_{spin} ; CP from F .

7.7 Strong CP from Orientation $\mathbb{Z}_2$ (HCSM-43)

Theorem 6. $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$ exactly.

On the 14-mode multiplet at $\lambda = 4$, the diagonal flip $F : (n_1, n_2) \rightarrow (n_2, n_1)$ satisfies $[F, \Delta] = 0$, $\{F, H_{\text{wind}}\} = 0$ exactly (verified at machine precision), and $[F, H_{\text{wind}}] \neq 0$. The winding matrix M is antisymmetric under F ; the odd function $\sin(\pi M/8)$ preserves the antisymmetry; hence $P_F H_{\text{wind}} P_F^{-1} = -H_{\text{wind}}$. The physical identification $H_{\text{wind}} \leftrightarrow F_{\mu\nu} \tilde{F}^{\mu\nu}$ gives $\theta_{\text{eff}} = 0$ exactly, because H_{wind} is F -odd and the substrate vacuum is an F -eigensector.

7.8 Proca Sector and $\mathfrak{u}(1)_Y$ Origin (HCSM-44)

The $\mathfrak{u}(1)_Y$ gauge symmetry is generated by the hypercharge functional $\hat{Y}$ on $\Omega^0|_{\lambda=4}$: $[\hat{Y}, T] = 0$, $[\hat{Y}, F] = 0$, eigenvalues in $(1/6)\mathbb{Z}$. The Proca Lagrangian follows from the HCSM Higgs mechanism:

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{1}{2}m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu, \quad (5)$$

with $m_W = gv_{\text{EW}}/2$, $m_Z = \sqrt{g^2 + g'^2}v_{\text{EW}}/2$. The mass ratio $m_W/m_Z = \cos\theta_W = 0.8769$, matching PDG 0.8814 at 0.5%. Tree-level $\rho = 1$ exactly.

8 Level 6 — Cosmology

8.1 The kL crossover

$kL_{\text{bare}} = (d-1)N/(d+1) = 192/5 = 38.4$ exact; $kL_{\text{phys}} = 38.442527$, matching $\ln(M_P/v_{\text{EW}})$ to 0.004%. Correction: $kL_{\text{phys}} - kL_{\text{bare}} = (|D_4| - 1)/2 \cdot \delta_{\text{edge}} = (7/2)\delta_{\text{edge}}$.

8.2 Dark energy and the DME entropy-well normalization

Bare formula: $\rho_{\text{DE}} = (|F|/|B|)m_\nu^4 = (d+1)/2 \cdot m_\nu^4 = 2.4809 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.76%.

Refined formula (HCSM-52, HCSM-53):

$$\rho_{\text{DE}} = \frac{d+1}{2} (1 + c \cdot \delta) m_\nu^4, \quad c = \frac{d+1}{(d-1)^2} = \frac{5}{9}, \quad \delta = \frac{(1-\gamma)^2}{2\pi}, \quad (6)$$

giving $\rho_{\text{DE}} = 2.5003 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.010%.

$S_0 = v_{\text{EW}}^2 \cdot 7/40 \cdot (1 + \delta_{\text{edge}}/6) = 10630.722 \text{ GeV}^2$, matching the implied value at relative residual 2.44×10^{-6} (18.8% of the M_P budget). Single-anchor corollary: M_P is the only dimensionful input.

8.3 Vacuum stability

$\gamma = c_2/I(2) = 0.702839467990\dots$, with $I(2) = 1 - \frac{1}{2} \text{AGM}(1, \sqrt{2})$, $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$, $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. Stability theorem: $S''(0)/S_0 = -2\gamma < 0$; the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy.

8.4 The Λ -promotion mechanism

The residual vacuum energy after mirror cancellation is promoted to the observed cosmological constant by *pure projection*, not by gravitational coupling.

The five-layer loop $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ acts on the 14-mode multiplet. The EFT layer L_{EFT} acts as a *soft projector* with weight $1 + c\delta$, where $c = 5/9$ is set by the kernel-diagonal theorem (Section 5.10). The residual energy projects onto the timelike direction $(v_0 + v_2)/\sqrt{2}$ of $\ker D$, with unit projection weight (the residual is a scalar). The hidden $\mathfrak{u}(1)$ factors contribute zero: their generators are traceless on each D_4 multiplicity space of the hidden sector, and the hidden vacuum is the D_4 -invariant nullspace of the hidden measurement matrix, so all hidden $\mathfrak{u}(1)$ VEVs vanish at machine precision.

The mechanism is therefore a projection, not a coupling. There is no gravitational suppression factor $1/M_P^2$; the residual vacuum energy is already at the right scale, because it is topological. Combined with the absence of a light dark photon (HCSM-41), this confirms that no gauge mediator exists between the hidden and visible sectors, and the only route for the residual energy to affect the visible sector is universal gravitation via the projection onto $\ker D$. This closes item 6.13.

9 Session-Level Theorem Inventory

The following theorems are established in the HCSM framework and are integrated throughout this paper:

Theorem	Location
$N = d \cdot 2^d$	HCSM-01
$\dim \ker D = 4$; emergent signature $(3, 1)$	HCSM-02
$\mathcal{O}_2 = \text{Fix}(M)$; $\mathbb{Z}_2 \times \mathbb{Z}_2$ fixed points $= \{W, t\}$	HCSM-05
Winding generator spectrum	HCSM-14
Class partition $(3, 7, 4)$	HCSM-14
$\{H_{\text{wind}}, P_{rs}\} = 0$	HCSM-14, HCSM-43
Hartree orbit trace $(2, 3, 2)$, ratio $3/2$ exact	HCSM-19
$T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma)$	HCSM-21
Zero state s ; rank 13, nullity 1	HCSM-22
Charge bijection $q_3 = F(n_1, \sigma)$	HCSM-24
Observer pair $\{W, t\}$	HCSM-26
Top-Yukawa $4/3 = \mathcal{O}_2 /(d-1)$; $m_t = 172.688 \text{ GeV}$	HCSM-40

Theorem	Location
$\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$	HCSM-40
Bridge identity at 8.4×10^{-8}	HCSM-40
Topological protection of $\phi_{\text{gen}} = (3/5)\pi$	HCSM-37
$\epsilon = 0$ exactly (no dark photon)	HCSM-41
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ on 56-dim	HCSM-42
$\{\gamma_5, D\} = 0$; two-loop anomaly cancellation	HCSM-42
Strong CP: $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$	HCSM-43
$u(1)_Y$ origin; Proca; $m_W/m_Z = \cos \theta_W$	HCSM-44
Graviton masslessness by Schur	HCSM-45, 46
$G_N = (d+1)^2/(8\pi M_P^2)$	HCSM-48
kL correction $(7/2)\delta_{\text{edge}}$	HCSM-51
EFT coefficient $c = (d+1)/(d-1)^2 = 5/9$	HCSM-52
$S_0 = v_{\text{EW}}^2 \cdot 7/40 \cdot (1 + \delta_{\text{edge}}/6)$	HCSM-53
Vacuum stability $\gamma = c_2/I(2)$	HCSM-54
Kernel-Diagonal Theorem	HCSM-60
Three coefficients from one diagonal	HCSM-60
F-invariance of the winding kernel	HCSM-60
Kernel-T decomposition	HCSM-60
Λ-promotion is a projection (item 6.13 closed)	Section 8.4

All seven rank blockers of the framework’s derivation program are closed:

Rank	Blocker	Status	Paper
1	HCSM-native γ	Closed	HCSM-54
2	kL bare	Closed	HCSM-51
3	Charge bijection	Closed	HCSM-24
4	$N = d \cdot 2^d$	Closed	HCSM-01
5	$(d-1)/2$ from DNLS	Closed	HCSM-19
6	3D irrep for CKM	Closed by reframing	HCSM-34
7	Loop-pass duration	Closed	HCSM-21

10 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions. The verification scripts are indexed by paper number; running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the stack.

The complete HCSM paper stack, the verification scripts, and the LaTeX and TXT source of each paper are available at <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->.

11 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by two independent anchors (photon helicity for $d = 4$; equal entropy spacing for $L = 8$) and a derived third anchor (the topology of the Hodge-Dirac kernel). The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer loop are the machine’s operation. The output structures — charge, mass, flavor, gravity, cosmology — are the residue after mirror cancellation.

The framework’s disclosed-input set consists of three discrete inputs (Standard Model content; physical identifications; identification thesis) and one dimensionful anchor (the Planck mass M_P). Every other quantity is derived from the substrate. All seven rank blockers are closed.

The session’s structural additions are three. First, the *Kernel-Diagonal Theorem* (HCSM-60) unifies the framework’s three most consequential coefficients under a single projector: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ are three traces of the diagonal of the projector onto $\ker(H_{\text{wind}})$. Second, the Λ -promotion mechanism (item 6.13) is closed: the residual vacuum energy is promoted by pure projection onto the timelike direction of $\ker D$, and the hidden $\mathfrak{u}(1)$ VEVs vanish. Third, the session’s supplementary results — absence of a light dark photon, spin-chirality $\mathbb{Z}_2$ grading, strong-CP solution, Proca sector, and the topological protection of ϕ_{gen} — complete the framework’s coverage of the Standard Model gauge content.

The framework’s claim is that the Standard Model is what the substrate looks like when you observe it. The Hodge complex is the machine. Physics is what the machine cannot cancel.

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